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PAGE NO.
DATE: / /

Q. Given three integer m, a & b, return mth magical no. since the ans may be very large return.

A magical no. if no. is divisible by either a or b.

m = 1 , a = 2, b = 3
out = 2.

⇒ Approach:-

i> Creating a modulo with value 10^9+7.

ii> Calculating lcm of a & b with the help of GCD. relation.

$$lcm = \frac{(a \times b)}{gcd(a,b)}$$

iii> Performing binary search

(start) s = min(a, b).

(end) e = m × min(a,b).

$$mid = \frac{(s+e)}{2}.$$

iv> Counting magical number ≤ mid.

$$count = \frac{mid}{a} + \frac{mid}{b} - \frac{mid}{lcm}.$$

if count < m then s = mid + 1

else ⇒ e = mid.

v> return lcm % MOD.

PAGE NO.
DATE: / /

Pseudocode

```
long gcd ( long a , long b){
        if(b==0) return a;
            return gcd (b, a%b);
}
long lcm (long a, long b){
            return ( a / gcd(a,b) ) * b;

int nth Magical Number (int n, int a , int b){
        int mod = 1000000001 ;
        long low = min(a,b);
            long high = n * min(a,b);
        long long L = lcm(a,b);
        while(low < high){
            long mid = low + (high-low)/2;
            long count = mid/a + mid/b - mid/c ;
            if(count<n)
                low = mid+1;
            else
                high = mid;
        } return low % mod;
}
int main (){
        System.out.println (nthMagicalNumber(1,2,3));
}
```

Output ⇒ 2.